THREE DIMENSIONAL GEOMETRY

WORKEDOUT EXAMPLES

W.E-1: The vector along the line of greatest slope in the plane $x+y+z=3$ with respect to the plane $x-y+z=0$ is

Ans: $-\hat{i}+2\hat{j}-\hat{k}$.

Sol: $\bar{a}=\hat{i}+\hat{j}+\hat{k}, b=\hat{i}-\hat{j}+\hat{k}$

$$\bar{a}\times\bar{b}=\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}=2\hat{i}-2\hat{k}=2\left(\hat{i}-\hat{k}\right)$$

$$\bar{a}\times\left(\bar{a}\times\bar{b}\right)=\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix}=-\hat{i}+2\hat{j}-\hat{k}$$

hence the vector along the line with greatest slope is $-\hat{i}+2\hat{j}-\hat{k}$

W.E-2. If the plane $ax+by=0$ is rotated through an angle 'α' about it's line of intersection with the plane z=0. Then the equation of the plane in the new position will be

A) $ax+by+z\sqrt{a^2+b^2}\tan\alpha=0$

B) $ax+by-z\sqrt{a^2+b^2}\tan\alpha=0$

C) $bx+ay+z\sqrt{a^2+b^2}\tan\alpha=0$

D) $bx+ay-z\sqrt{a^2+b^2}\tan\alpha=0$

Sol: (BCD)
Equation of the plane passing through the intersection of $ax+by=0$ and $z=0$ is

$ax+by+\lambda z=0$ --------(1)

Dcs of normal to the plane are

$$\frac{a}{\sqrt{a^2+b^2+\lambda^2}},\frac{b}{\sqrt{a^2+b^2+\lambda^2}},\frac{\lambda}{\sqrt{a^2+b^2+\lambda^2}}$$

$$\therefore \cos\alpha=\frac{a^2+b^2}{\sqrt{a^2+b^2+\lambda^2}\sqrt{a^2+b^2}}$$

$$\Rightarrow \lambda=\pm\sqrt{a^2+b^2}\tan\alpha$$

W.E-3. A rod of length 2 units whose one end is (1,0,-1) and the other lies on the plane $x-2y+2z+4=0$, then

A) The rod sweeps a figure of volume π cubic units

B) The area of the region which the rod traces on the plane is 2π square units

C) The length of the projection of the rod on the plane is $\sqrt{3}$ units

D) The centre of the region which the rod traces on the plane is $\left(\frac{2}{3},\frac{2}{3},\frac{-5}{3}\right)$

Sol. (ACD)
The rod sweeps a cone of height 1 and radius $\sqrt{3}$ units

W.E-4. Consider a set of points R in the space which are at a distance of 2 units from the line $\frac{x}{1}=\frac{y-1}{-1}=\frac{z+2}{2}$ between the planes $x-y+2z+3=0$ and $x-y+2z-2=0$ then

(IIT-JEE-2014)

A) The volume of the bounded figure by the points R and the planes is $\frac{10\pi}{3\sqrt{3}}$ cubic units

B) The area of the curved surface formed by the set of points R is $\frac{20\pi}{\sqrt{6}}$ square units

C) The volume of the bounded figure by the set of points in R and the planes is $\frac{20\pi}{\sqrt{6}}$ cubic units

D) The area of the curved surface formed by the set of points in R is $\frac{10\pi}{\sqrt{3}}$ square units

Sol: (BC)

The distance between the planes is $h=\frac{5}{\sqrt{6}}$

Also the figure formed is a cylinder, whose radius is $r=2units$

$$Hence\ the\ volume=\pi r^2h=\frac{20\pi}{\sqrt{6}}\ cubic\ units$$

$$Area of\ curved\ surface\ 2\pi rh=\frac{20\pi}{\sqrt{6}}\ square\ units$$

W.E-5. From a point $P(\lambda,\lambda,\lambda)$, perpendiculars PQ

and PR are drawn respectively on the lines $y=x, z=1$ and $y=-x, z=-1$. If P is such that $\angle QPR$ is a right angle, then the possible value(s) of λ is (are)

A) $\sqrt{2}$ B) 1 C) -1 D) $-\sqrt{2}$

sol: (C)

Let $Q=(k,k,1)$, $R=(l,-l,-1)$

$PQ \perp lar\ to\ \frac{x}{1}=\frac{y}{1}=\frac{z-1}{0}$

$\Rightarrow \lambda = k \Rightarrow Q=(\lambda,\lambda,1)$

$PR \perp lar\ to\ \frac{x}{-1}=\frac{y}{1}=\frac{z+1}{0}$

$\Rightarrow l=0 \Rightarrow R=(0,0,-1)$

Given $\angle QPR = 90^0$

$\Rightarrow PQ \perp PR \Rightarrow (\lambda-1)(\lambda+1)=0$

$\Rightarrow \lambda = 1,-1$

if $\lambda=1$ then P=Q which is not possible.

Hence $\lambda=-1$

W.E-6. Supose in a tetrahedron $ABCD$, $AB=1;\ CD\ \sqrt{3};$ the distance and angle between the skew lines AB and CD are 2 and $\frac{\pi}{3}$ respectively. If the volume of the tetrahedron is V then the values of $6V$ is

sol:

$|\vec{b}|=1 \left|\vec{c}-\vec{d}\right|=\sqrt{3}$

shortest distence between AB and CD is 2

also $\left(\vec{b},\vec{c}-\vec{d}\right)=\frac{\pi}{3}$

Let equation of A B be $\vec{r}=\lambda\vec{b}$ ......(1)

and equation of CD be $\vec{r}=\vec{c}+\mu\left(\vec{c}-\vec{d}\right)$ ....(2)

$\vec{n}=\vec{b}\times\left(\vec{c}-\vec{d}\right)$

$$S.D=\left|projection\ of\ \vec{c}\ on\ \hat{n}\right|=\left|\frac{\vec{c}.\hat{n}}{|\hat{n}|}\right|=\left|\frac{\vec{c}.\left(\vec{b}\times\left(\vec{c}-\vec{d}\right)\right)}{\vec{b}\left(\vec{c}-\vec{d}\right)}\right|$$

$$=\left|\frac{[\vec{c}\ \vec{b}\ \vec{c}]+[\vec{c}\ \vec{b}\ \vec{d}]}{|\vec{b}|\left|\vec{c}-\vec{d}\right|\sin(\pi/3)}\right|$$

$$2=\left|\frac{0+[\vec{b}\ \vec{c}\ \vec{d}]}{1.\sqrt{3}.\left(\sqrt{3}/2\right)}\right|$$

$$\left[\vec{b}\ \vec{c}\ \vec{d}\right]=3$$

Volume $=\frac{1}{6}[\vec{b}\ \vec{c}\ \vec{c}]=\frac{1}{6}.3=\frac{1}{2}$

W.E-7. Find the shortest distance of plane parallel to z-axis and containing line

$x+y+2z-3=0=2x+3y+4z-4$ from z-axis.

Ans. (2)

sol: Equation of line (1) is $\frac{x-5}{-2}=\frac{y+2}{0}=\frac{z}{1}$

Equation of line (2) is $\frac{x}{0}=\frac{y}{0}=\frac{z}{1}$

Shortest distance$=\left|\frac{\left(\bar{a}-\bar{c}\right).\bar{b}\times\bar{d}}{\bar{b}\times\bar{d}}\right|=\frac{4}{2}=2\ units$

W.E-8. **Statement I:** The plane $5x+2z-8=0$ contains the line $2x-y+z-3=0$ and $3x+y+z=5$ and is perpendicular to $2x-y-5z-3=0$

Statement II: The plane $3x+y+z=5$ meets the line $x-1=y+1=z-1$ at the point (1,1,1).

Sol: Equation of plane is

$2x-y+z-3+\lambda(3x+y+z-5)=0$

For $\lambda=1$, we get

5x+2z-8=0 which is perpendicular to

2x-y-5z-3=0 as $5\times2+0(-1)+2(-5)=0$

W.E-9. The reflection of the plane

$2x-3y+4z-3=0$ in the plane

$x-y+z-3=0$ is the plane.

A) $4x-3y+2z-15=0$

B) $x-3y+2z-15=0$

C) $4x+3y-2z+15=0$

D) $x-3y+2z-5=0$

Sol: (A)

$$2\left(aa'+bb'+cc'\right)\left(ax+by+cz+d\right)$$

$$=\left(a^2+b^2+c^2\right)\left(a'x+b'y+c'z+d'\right)$$

$\therefore$ Reflection of plane $2x-3y+4z-3=0$

in the plane $x-y+z-3=0$

$$2(2+3+4)(x-y-z-3)=3(2x-3y+4z-3)$$

$$\Rightarrow 4x-3y+2z-15=0$$

W.E-10. P,Q,R,S are four coplanar points on the sides AB,BC,CD,DA of a skew quadrilaterial. The product $\frac{AP}{PB}\cdot\frac{BQ}{QC}\cdot\frac{CR}{RD}\cdot\frac{DS}{SA}$ equals

A)-2 B)-1 C)2 D)1

sol: (D)

Let the vertices A,B,C,D of a quadrilateral be $(x_1,y_1,z_1),(x_2,y_2,z_2),(x_3,y_3,z_3),(x_4,y_4,z_4)$. The equation of plane PQRS be

$$u\equiv ax+by+cz+d=0$$

Let $u_r\equiv a_1x+b_1y+c_1z+d$

where $r=1,2,3,4$

Then $\frac{AP}{PB}\cdot\frac{BQ}{QC}\cdot\frac{CR}{RD}\cdot\frac{DS}{SA}$

$$=\left(-\frac{u_1}{u_2}\right)\left(-\frac{u_2}{u_3}\right)\left(-\frac{u_3}{u_4}\right)\left(-\frac{u_4}{u_1}\right)=1$$

W.E-11. Perpendiculars are drawn from points on the line $\frac{x+2}{2}=\frac{y+1}{-1}=\frac{z}{3}$ to the plane $x+y+z=3$. The feet of perpendiculars lie on the line. **(IIT JEE-2013)**

A) $\frac{x}{5}=\frac{y-1}{8}=\frac{z-2}{-13}$ B) $\frac{x}{2}=\frac{y-1}{3}=\frac{z-2}{-5}$

C) $\frac{x}{4}=\frac{y-1}{3}=\frac{z-2}{-7}$ D) $\frac{x}{2}=\frac{y-1}{-7}=\frac{z-2}{5}$

Sol: (D)

Any point B on line is $(2\lambda-2,-\lambda-1,3\lambda)$

Point B lies on the plane for some 'λ'

$$\Rightarrow(2\lambda-2)+(-\lambda-1)+3\lambda=3$$

$$\Rightarrow 4\lambda=6\Rightarrow\lambda=\frac{3}{2}\Rightarrow B=\left(1,\frac{-5}{2},\frac{9}{2}\right)$$

The foot of the perpendicular from point (-2,-1,0) on the plane is the point A(0,1,2)

$$D.r's\ of\ AB=\left(1-\frac{7}{2},\frac{5}{2}\right)=(2,-7,5)$$

Hence the required line is $\frac{x}{2}=\frac{y-1}{-7}=\frac{z-2}{5}$

W.E-12. Consider the lines

$$L_1:\frac{x-1}{2}=\frac{y}{-1}=\frac{z+3}{1},\ L_2:\frac{x-4}{1}=\frac{y+3}{1}=\frac{z+3}{2}$$

and the planes $P_1:7x+y+2z=3$, $P_2:3x+5y-6z=4$. Let $ax+by+cz=d$ be the equation of the plane passing through the point of intersection of lines L_1 and L_2 and perpendicular to planes P_1 & P_2 match ListI with ListII and select the correct answer using the code given below the lists. **(IIT JEE 2013)**

	List-I		List-II
P)	a=	1)	13
Q)	b=	2)	-3
R)	c=	3)	1
S)	d=	4)	-2

codes:

P **Q** **R** **S**

A	3	2	4	1
B	1	3	4	2
C	3	2	1	4
D	2	4	1	3

Sol: (A)

Plane perpendicular to P_1 and P_2 has D.r's of

normal $\begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 7 & 1 & 2 \\ 3 & 5 & -6 \end{vmatrix} = -16\bar{i} + 48\bar{j} + 32\bar{k}$.....(1)

For point of intersection of lines

$(2\lambda_1 + 1, -\lambda, \lambda_1 - 3) = (\lambda_2 + 4, \lambda_2 - 3, 2\lambda_2 - 3)$

$\Rightarrow 2\lambda_1 + 1 = \lambda_2 + 4\ (or)\ 2\lambda_1 - \lambda_2 = 3$

$\Rightarrow -\lambda_1 = \lambda_2 - 3\ (or)\ \lambda_1 + \lambda_2 = 3$

$\Rightarrow \lambda_1 = 2, \lambda_2 = 1$

$\therefore$ Point is (5,-2,-1) ....(2)

From (1) and (2) required plane is

$-1(x-5) + 3(y+2) + 2(z+1) = 0$

or $x - 3y - 2z = 13$

$\Rightarrow a = 1, b = -3, c = -2, d = 13$

W.E-13. Let $A = (-4,0,3), B = (14,2,-5)$ then which of the following points lie on the bisector of the angle between $\overline{OA}$ and $\overline{OB}$ (Where 'O' is the orgin of reference) is(are)

A) (2, 1, -1) B) (2, 11, 5)

C) (10, 2, -2) D) (1, 1, 2)

Sol : (D) $\overline{OA} = -4\hat{i} + 3\hat{k}$

$\overline{OB} = 14\hat{i} + 2\hat{j} - 5\hat{k}$

$\hat{OA} = \dfrac{-4\hat{i} + 3\hat{k}}{5}, \hat{OB} = \dfrac{14\hat{i} + 2\hat{j} - 5\hat{k}}{15},$

Any vector in the direction of angle bisectors of the from $\lambda\left(\hat{a} + \hat{b}\right)$

$= \dfrac{\lambda}{15}\left(2\hat{i} + 2\hat{j} + 4\hat{k}\right)$

$\therefore$ Answer is D with $\lambda = \dfrac{15}{2}$

Passage:

Two lines whose equations are $\dfrac{x-3}{2} = \dfrac{y-2}{3} = \dfrac{z-1}{\lambda}$

and $\dfrac{x-2}{3} = \dfrac{y-3}{2} = \dfrac{z-2}{3}$ lie in the same plane then

W.E-14. The value of $\sin^{-1}(\sin\lambda)$ is equal to

A) 3 B) $\pi - 3$ C) 4 D) $\pi - 4$

W.E-15. Point of intersection of the lines lie on

A) $3x + y + z = 20$ B) $3x + y + z = 25$

C) $3x + 2y + z = 24$ D) $3x + 2y - z = 10$

Sol.

14 (d)

Both lines are coplanar

$\Rightarrow \begin{vmatrix} 2 & 3 & \lambda \\ 3 & 2 & 3 \\ 1 & -1 & -1 \end{vmatrix} = 0$

$\Rightarrow 2(-2+3) + 3(3+3) + \lambda(-3-2) = 0$

$\Rightarrow \lambda = 4$

$\sin^{-1}(\sin 4) = \sin^{-1}\sin(\pi - 4) = \pi - 4$

15 (b) Let $\dfrac{x-3}{2} = \dfrac{y-2}{3} = \dfrac{z-1}{4} = r_1$

$\Rightarrow x = 3 + 2r_1, y = 3r_1 + 2, z = 4r_1 + 1$

it will lie on $\dfrac{x-2}{3} = \dfrac{y-3}{2} = \dfrac{z-2}{3} \Rightarrow r_1 = 1$

So, point of intersection is (5,5,5)

LEVEL - V

SINGLE ANSWER QUESTIONS

1. **P(0, 5, 6), Q(1, 4, 7), R(2, 3, 7) and S(3, 4, 6) are four points in the space. The point nearest to the origin O(0, 0, 0) is**
(A)only P (B)only Q
(C)R & S (D)P & S

2. **Four points** $A(+1,-1,1); B(1,3,1); C(4,3,1)$ **and** $D(4,-1,1)$ **taken in order are the vertices of**
(A)a parallelogram which is neither a rectangle nor a rhombus
(B) rhombus
(C)an isosceles trapezium
(D) a cyclic quadrilateral.

3. **The point in which the YZ plane divides the line joining the points (3, 5, –7) and (–2, 1, 8) is (x, y, z). Then the value of x + 5y + z is**
(A)10 (B)15 (C)12 (D)20

4. **P(1, 1, 1) and Q(l, l, l) are two points in the space such that** $PQ=\sqrt{27}$ **, the value of l can be**
(A)–4 (B)–2 (C)2 (D)0

5. **The plane** $\frac{x}{1}+\frac{y}{2}+\frac{z}{-1}=1$ **meets the co-ordinate axes in the points A, B and C respectively. Area of triangle ABC is**
(A) $\frac{3}{2}$ (B) $\frac{2\sqrt{2}}{3}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{6}}{2}$

6. **The direction ratios of the bisector of the angle between the lines whose** $l_1, m_1, n_1;\ l_2, m_2, n_2$ **are**
(A) $l_1 \pm l_2, m_1 \pm m_2, n_1 \pm n_2$
(B) $l_1^2+l_2^2, m_1^2+m_2^2, n_1^2+n_2^2$
(C) $l_1m_2-l_2m_1,\ m_1n_2-m_2n_1,\ n_1l_2-n_2l_1$
(D) $l_1m_2+l_2m_1,\ m_1n_2+m_2n_1,\ n_1l_2+n_2l_1$

7. **The plane which contains the line 3x + y = 1, z = 4 and parallel to x + y + z + 1 = 0, y + 2z = 1, cuts the x-axis at**
(A)(–2, 0, 0) (B)(–3, 0, 0)
(C)(–4, 0, 0) (D)(–1, 0, 0)

8. **If a straight line is given by** $\vec{r}=(1+t)\hat{i}+3t\hat{j}+(1-t)\hat{k}$ **where** $t\in R$.
If this line lies in the plane $x+y+cz=d$ **then the value of c + d is**
(A)-1 (B) 1 (C)7 (D) 9

9. **The line** $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ **and the plane 2x –4y + 2 z = 3 meet in**
(A)at one point (B)no point
(C)infinitely many points
(D)at two points

10. **If** $(4k_1, k_1^2, 1)$ and $(4k_2, k_2^2, 1)$ **are two points lying on the plane in which (2, 3, 2) and (1, 2, 1) are mirror image to each other, then** k_1k_2 **is equal to**
(A) $-\frac{3}{2}$ (B) $-\frac{5}{2}$ (C) $-\frac{7}{2}$ (D) $-\frac{9}{2}$

11. **The plane** $x-y-z=2$ **is rotated through an angle** 90^0 **about its line of intersection with the plane** $x+2y+z=2$**. Then equation of this plane in new position is**
(A) $5x+4y+z-10=0$
(B) $4x+5y-3z=0$
(C) $2x+y+2z=9$
(D) $3x+4y-5z=9$

12. **The ratio in which the join of (1, –2, 3) and (4, 2, –1) is divided by the XOY plane is**
(A) 1 : 3 B) 3 : 1 C) - 3: 1 D) - 4 : 1

13. **Let** a_i **, i = 1, 2, 3, ..., n denote the integers in the domain of function**
$$f(x)=\sqrt{\log_{\frac{1}{2}}\left(\frac{4x-25}{x-21}\right)}$$
where $a_i < a_{i+1}$, $i\in\mathbb{N}$. **If the line**
$$L: \frac{2x-a_1}{4}=\frac{y+a_1}{a_2}=\frac{z-a_3}{a_5}$$ **meets the xy,**

yz and zx planes at A, B and C respectively, and if volume of the tetrahedron OABD is V, where 'O' is origin and D is the image of C in the x-axis, then the value of 90V is

(A) 240 (B) 260 (C) 280 (D) 300

14. **Given a tetrahedron DABC with $AB = 12$, $CD = 6$. If the shortest distance between the skew lines AB and CD is 8 and the angle between them is $\frac{\pi}{6}$, then the volume of tetrahedron (in cubic units) is**

(A) 12 (B) 36 (C) 48 (D) 72

15. **The value of k such that** $\frac{x-4}{1}=\frac{y-2}{1}=\frac{z-k}{2}$ **lies in the plane** $2x-4y+z=7$**, is** **[IIT-JEE 2003]**

(A) 7 (B) -7 (C) no real value (D) 4

16. **Let** $P(3,2,6)$ **be a point in space and Q be a point on the line** $\vec{r}=(\hat{i}-\hat{j}+2\hat{k})+\mu(-3\hat{i}+\hat{j}+5\hat{k})$. **Then the value of** μ **for which the vector** $\overrightarrow{PQ}$ **is parallel to the plane** $x-4y+3z=1$ **is** **[IIT-JEE 2009]**

(A) $\frac{1}{4}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $-\frac{1}{8}$

17. **A line with positive direction cosines passes through the point P(2,-1,2) and makes equal angles with the coordinate axes. The line meets the plane 2x + y + z = 9 at point Q. The length of the line segment PQ equals** **[IIT-JEE 2009]**

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

18. **A plane passes through (1,-2,1) and is perpendicular to two planes** $2x-2y+z=0$ **and** $x-y+2z=4$**, then the distance of the plane from the point (1,2,2) is** **[IIT-2006]**

(A) 0 (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

19. **If** α,β,γ **be the angles which a line makes with the coordinate axes, then**

(A) $\sin^2\alpha=\cos^2\beta+\cos^2\gamma$

(B) $\cos^2\alpha=\cos^2\beta+\cos^2\gamma$

(C) $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

(D) $\sin^2\alpha+\sin^2\beta=1+\cos^2\gamma$

20. **The direction cosines of a line equally inclined with the coordinate axes are**

(A) $1/\sqrt{3},\ -1/\sqrt{3},\ 1/\sqrt{3}$

(B) $1/\sqrt{3},\ -1/\sqrt{3},\ -1/\sqrt{3}$

(C) $-1/\sqrt{3},\ -1/\sqrt{3},\ 1/\sqrt{3}$

(D) $1/\sqrt{3},\ 1/\sqrt{3},\ 1/\sqrt{3}$

21. **If the direction cosines l, m, n of a line are related by the equations l+m+n=0, 2mn+2ml- nl=0 then the ordered triplet (l, m, n) is**

(A) $\left(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{-2}{\sqrt{6}}\right)$

(B) $\left(-\frac{1}{\sqrt{6}},-\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}\right)$

(C) $\left(-\frac{2}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}}\right)$

(D) $\left(\frac{2}{\sqrt{6}},-\frac{1}{\sqrt{6}},-\frac{1}{\sqrt{6}}\right)$

22. **The lines** $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ **and** $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ **are coplanar if**

(A) $k=0$ (B) $k=-1$ (C) $k=-3$ (D) $k=3$

23. **The lines** $\frac{x-1}{3}=\frac{y-1}{-1}=\frac{z+1}{0}$ **and** $\frac{x-4}{2}=\frac{y+0}{0}=\frac{z+1}{3}$

(A) do not intersect

(B) intersect
(C) intersect at (4, 0, –1)
(D)intersect at (1, 1, –1)

24. The line $\frac{x-1}{1}=\frac{y-2}{-2}=\frac{z-1}{3}$ **and the plane** $x+2y+z=6$ **meet in**
(A) no point
(B) the line lies on the plane
(C) infinitely many points
(D) at one point

25. The plane $x-2y+7z+21=0$
(A) contains the line $\frac{x+1}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$
(B) contains the point (0, 7, –1)
(C) is perpendicular to the line $\frac{x}{1}=\frac{y}{-2}=\frac{z}{7}$
(D) is parallel to the plane $x-2y+7z=0$

26. The equation of a line $4x-4y-z+11=0=x+2y-z-1$ **can be put as**
(A) $\frac{x}{2}=\frac{y-2}{1}=\frac{z-3}{4}$
(B) $\frac{x-4}{2}=\frac{y-4}{1}=\frac{z-11}{4}$
(C) $\frac{x-2}{2}=\frac{y}{1}=\frac{z-3}{4}$
(D) $\frac{x-2}{2}=\frac{y-2}{1}=\frac{z}{4}$

27. If a straight line makes an angle of 60° with each of the x and y-axes, the angle which it makes with the z-axis is
(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

28. The lines $\vec{r}=\hat{i}+\hat{j}-\hat{k}+\lambda(3\hat{i}-\hat{j})$, and $\vec{r}=(4\hat{i}-k)+\mu(2\hat{i}+3\hat{k})$
(A) meet in a unique point (B) are skew lines
(C) are coplanar (D)are coincident

29. If the plane 2x - 3y + 6z - 11 = 0 makes an angle θ **with the x-axis then**
(A) $\cos\theta=\frac{2}{7}$ (B) $\sin\theta=\frac{2}{7}$
(C) $\tan\theta=\frac{\sqrt{2}}{3}$ (D) $\tan\theta=\frac{2}{3\sqrt{5}}$

30. The angle θ **between the line** $\vec{r}=\vec{a}+\lambda\vec{b}$ *and* $\vec{r}.\vec{n}=p$ *is*
(A) $\sin^{-1}\left(\frac{\vec{b}.\hat{n}}{|\vec{b}|}\right)$ (B) $\cot^{-1}\left(\frac{\vec{b}.\hat{n}}{|\vec{b}|}\right)$
(C) $\frac{\pi}{2}-\cos^{-1}\left(\frac{\vec{b}.\hat{n}}{|\vec{b}|}\right)$ (D) $\cos^{-1}\left(\frac{\vec{a}.\hat{n}}{|\vec{a}|}\right)$

31. If P(2, 3, 1) is a point and $L\equiv x-y-z-2=0$ **is a plane then**
(A) origin and P lie on the same side of the plane
(B)distance of P from the plane is $\frac{4}{\sqrt{3}}$
(C) foot of perpendicular is $\left(\frac{10}{3},\frac{5}{3},-\frac{1}{3}\right)$
(D) image of point P by the plane $\left(\frac{10}{3},\frac{5}{3},-\frac{1}{3}\right)$

32. Consider the plane through (2, 3, –1) and at right angles to the vector $3\hat{i}-4\hat{j}+7\hat{k}$ **from the origin is**
(A)The equation of the plane through the given point is $3x-4y+7z+13=0$
(B)perpendicular distance of plane from origin $\frac{1}{\sqrt{74}}$
(C)perpendicular distance of plane from origin $\frac{13}{\sqrt{74}}$
(D)perpendicular distance of plane from origin $\frac{21}{\sqrt{74}}$

33. Consider the point (1, 3, 4) and the plane $\vec{r}.(2\hat{i}-\hat{j}+\hat{k})+3=0$. **Then**
(A)The projection of the point in the plane is (1, 3, 4)
(B)The perpendicular distance from the point to the given plane is $\sqrt{6}$.

(C)The projection of the point in the plane is $(-1, 4, 3)$

(D)The projection of the point in the plane is $(-5, 4, 3)$

34. **A variable plane cutting coordinate axes in A,B,C is at a constant distance 1 from the origin. Then the locus of centroid of the triagle** ABC is

(A) $X^{-2}+Y^{-2}+Z^{-2}=16$

(B) $X^{-2}+Y^{-2}+Z^{-2}=9$

(C) $\frac{1}{9}\left\{\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right\}-1=0$

(D) $X+Y=0$

35. **A line L passing through the point P(1, 4, 3), is perpendicular to both the lines** $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$, **and** $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$. **If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then ($a_1 + a_2 + a_3$) can be**

(A) 16 (B) 15 (C) 2 (D) 1

36. **If OABC is a tetrahedron such that** $OA^2+BC^2=OB^2+CA^2=OC^2+AB^2$ **then**

(A) $OA \perp BC$ (B) $OB \perp CA$

(C) $OC \perp AB$ (D) $AB \perp BC$

37. **The base of the pyramid AOBC is an equilateral triangle OBC with each side is equal to** $4\sqrt{2}$. **'O' being the origin of reference, AO is perpendicular to the plane of triangle OBC and AO = 2. Then the cosine of the angle between the skew lines one passes through A and midpoint of OB, and the other passing through O and the mid point of BC can be**

(A) $-\frac{1}{\sqrt{2}}$ (B) 0 (C) $\frac{1}{\sqrt{6}}$ (D) $\frac{1}{\sqrt{2}}$

COMPREHENSION QUESTIONS

PASSAGE - I

Eq. of line is $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$

Equation of plane through the intersection of two planes is

$$\left(a_1x+b_1y+c_1z+d_1\right)+\lambda\left(a_2x+b_2y+c_2z+d_2\right)=0$$

38. **The distance of point (1, - 2, 3) from the plane** $x-y+z=5$ **measured parallel to the line** $\frac{x}{2}=\frac{y}{3}=\frac{z-3}{-4}$ **is**

(A) $\frac{\sqrt{21}}{5}$ (B) $\frac{\sqrt{29}}{5}$ (C) $\frac{\sqrt{13}}{5}$ (D) $\frac{2}{\sqrt{5}}$

39. **The equation of the plane through (0,2,4) and containing the line** $\frac{x+3}{3}=\frac{y-1}{4}=\frac{z-2}{-2}$ **is**

(A) $x-2y+4z-12=0$

(B) $5x+y+9z-38=0$

(C) $10x-12y-9z+60=0$

(D) $7x+5y-3z+2=0$

PASSAGE - II

$\vec{a}=6\hat{i}+7\hat{j}+7\hat{k}$, $\vec{b}=3\hat{i}+2\hat{j}-2\hat{k}$, $P(1,2,3)$

40. **The position vector of L, the foot of the perpendicular from P on the line** $\vec{r}=\vec{a}+\lambda\vec{b}$ **is**

(A) $6\hat{i}+7\hat{j}+7\hat{k}$ (B) $3\hat{i}+2\hat{j}-2\hat{k}$

(C) $3\hat{i}+5\hat{j}+9\hat{k}$ (D) $9\hat{i}+9\hat{j}+5\hat{k}$

41. **The image of the point P in the line** $\vec{r}=\vec{a}+\lambda\vec{b}$ **is**

(A) (11, 12, 11) (B) (5, 2, -7)

(C) (5, 8, 15) (D) (17, 16, 7)

42. **If A is the point with position vector** $\vec{a}$ **then area of the triangle** PLA **in sq. units is equal to**

(A) $3\sqrt{6}$ (B) $\frac{7\sqrt{17}}{2}$ (C) $\sqrt{17}$ (D) $\frac{7}{2}$

PASSAGE - III

A (-2, 2, 3) and B (13, -3, 13) and L is a line through A

43. A point P moves in the space such that 3PA = 2PB, then the locus of P is

(A) $x^2+y^2+z^2+28x-12y+10z-247=0$

(B) $x^2+y^2+z^2-28x+12y+10z-247=0$

(C) $x^2+y^2+z^2+28x-12y-10z+247=0$

(D) $x^2+y^2+z^2-28x+12y-10z+247=0$

44. Coordinates of the point P which divides the join of A and B in the ratio 2 : 3 internally are

(A) $(33/5,-2/5,9)$

(B) (4, 0, 7)

(C) $(32/5,-12/5,17/5)$

(D)(20, 0, 35)

45. Equation of a line L, pependicular to the line AB is

(A) $\frac{x+2}{15}=\frac{y-2}{-5}=\frac{z-3}{10}$

(B) $\frac{x-2}{3}=\frac{y+2}{13}=\frac{z+3}{2}$

(C) $\frac{x+2}{3}=\frac{y-2}{13}=\frac{z-3}{2}$

(D) $\frac{x-2}{15}=\frac{y+2}{-5}=\frac{z+3}{10}$

PASSAGE - IV

Given points A (1, -4, 5) and B(0,6,1) and a plane $3x-y+2z=7$

46. The ratio in which the line segment AB is divided by the plane, is

(A)2/3 (B)1/11 (C)10/11 (D)12/11

47. If $P(\lambda^2+1,\lambda,\lambda-1)$ is a point on the same side of the plane as the point A, then the set of values of λ, is

(A) $\left(\frac{-1-\sqrt{73}}{6},\frac{-1+\sqrt{73}}{6}\right)$

(B) $\left(-\infty,\frac{-1-\sqrt{73}}{6}\right)\cup\left(\frac{-1+\sqrt{73}}{6},\infty\right)$

(C) $(-\infty,\infty)$ (D) $(0,\infty)$

PASSAGE – V

Read the following passage and answer the questions Consider the lines

$L_1:\frac{x+1}{3}=\frac{y+2}{1}=\frac{z+1}{2}$,

$L_2:\frac{x-2}{1}=\frac{y+2}{2}=\frac{z-3}{3}$

48. The unit vector perpendicular to both L_1 and L_2 is

(A) $\frac{-\hat{i}+7\hat{j}+7\hat{k}}{\sqrt{99}}$ (B) $\frac{-\hat{i}-7\hat{j}+5\hat{k}}{5\sqrt{3}}$

(C) $\frac{-\hat{i}+7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (D) $\frac{7\hat{i}-7\hat{j}-\hat{k}}{\sqrt{99}}$

49. The shortest distance between L_1 and L_2 is

(A) 0 unit (B) $\frac{17}{\sqrt{3}}$ unit

(C) $\frac{41}{5\sqrt{3}}$ unit (D) $\frac{17}{5\sqrt{3}}$ unit

50. The distance of the point $(1,1,1)$ from the plane passing through the point $(-1,-2,-1)$ and whose normal is perpendicular to both the lines L_1 and L_2 is [IIT-JEE 2008]

(A) $\frac{2}{\sqrt{75}}$ unit (B) $\frac{7}{\sqrt{75}}$ unit

(C) $\frac{13}{\sqrt{75}}$ unit (D) $\frac{23}{\sqrt{75}}$ unit

PASSAGE: VI

A plane P contains the line $L_1:\frac{y}{b}+\frac{z}{c}=1$, x = 0 and is parallel to the line $L_2:\frac{x}{a}-\frac{z}{c}=1$, y = 0.

51. Equation of the plane P is

(A) $\frac{x}{a}-\frac{y}{b}+\frac{z}{c}+1=0$ (B) $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}-1=0$

(C) $\frac{x}{a}+\frac{y}{b}-\frac{z}{c}-1=0$ (D) $\frac{x}{a}-\frac{y}{b}-\frac{z}{c}+1=0$

52. **If shortest distance between the lines L_1 and L_2 is $\frac{1}{4}$ then the value of $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$ equals**

(A) 16 (B) 64 (C) 128 (D) 192

53. **Distance of image of A(a, 0, 0) in the plane P from M $\left(\frac{-5}{3},\frac{8}{3},\frac{11}{3}\right)$, where $a=b=c=1$, is equal to**

(A) 1 (B) 2 (C) 3 (D) 4

PASSAGE: VII

Consider the line L : $\frac{x-1}{2}=\frac{y}{1}=\frac{z+1}{-2}$ and a point A(1, 1, 1). Let P be the foot of the perpendicular from A on L and Q be the image of the point A in the line L, 'O' being the origin.

54. **The distance of the origin from the plane passing through the point A and containing the line L is**

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

55. **The distance of the point A from the line L is**

(A) 1 (B) 2 (C) $\sqrt{3}$ (D) $\frac{4}{3}$

56. **The distance of the origin from the point Q is**

(A) $\sqrt{3}$ (B) $\sqrt{\frac{17}{6}}$ (C) $\sqrt{\frac{17}{3}}$ (D) $\frac{1}{\sqrt{3}}$

ASSERTION & REASON QUESTIONS

(A) Statement -1 is true, statement -2 is true, statement -2 is a correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true, statement -2 is not correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false

(D) Statement -1 is false, statement-2 is true

57. **Statement 1 :**

Lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ will intersect if $\left(\vec{b}_1\times\vec{b}_2\right).\left(\vec{a}_2-\vec{a}_1\right)=0$ because

Statement 2 : Two lines will intersect if the shortest distance between them is zero.

58. **Statement 1 : If P(–3, –2, 4), Q(–9, –8, 10) and R(–5, –4, 6) are given, then on joining P, Q and R a unique plane will be formed. because**

Statement 2 : If three points are given, then it is not always possible to form a unique plane.

59. **Statement -1 : Let θ be the angle between the line $\frac{x-2}{2}=\frac{y-1}{-3}=\frac{z+2}{-2}$ and the plane x+y-z=5 then $\theta=\sin^{-1}\left(-\frac{2}{\sqrt{51}}\right)$**

Statement - 2: Angle between a straight line and a plane is the complement of angle between the line and normal to the plane

60. **Consider the following statements:**

Statement 1 : The plane y+z+1=0 is parallel to x- axis.

Statement 2 : Normal to the plane is parallel to x- axis.

61. **Statement 1 : The point A(2, 9, 12), B(1, 8, 8), C(–2, 11, 8) and D(–1, 12, 12) are the vertices of a rhombus.**

Statement 2 : AB = BC = CD = DA and AC = BD

62. **Statement 1 : The points (2, 1, 5) and (3, 4, 3) lie on opposite side of the plane 2x + 2y – 2z – 1 = 0.**

Statement 2 : The algebraic perpendicular distance from the given points to the line have opposite sign.

63. **Consider the planes**

$3x-6y-2z=15$ **and** $2x+y-2z=5$

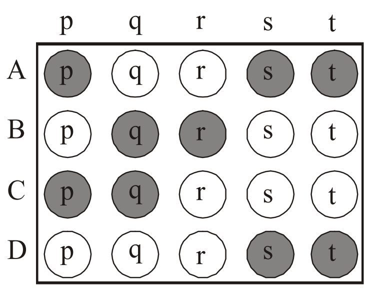
Statement – 1 : The parametric equation of the line of intersection of the given planes are x = 3 + 14t, y = 1 + 2t, z = 15t

Statement – 2 : The vector $14\hat{i}+2\hat{j}+15\hat{k}$ is parallel to the line of intersection of the given planes.

[IIT-JEE 2007]

MATRIX- MATCHING QUESTIONS

This section contains 1 questions. Each questions contain statements given in two columns, which have to be matched. The statements in **Column I** are labeled A, B, C and D while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example. If the correct matches are A-p, s and t, B-q and r, C-p and q, and D-s and t, then the correct darkening of bubbles will look like the following



64. Match the statements/expressions given in Column I with the values given in Column II

Column I

(A)The area of the triangle whose vertices are (0,0,0), (3,4,7) and (5,2,6) is

(B)Distance of plane $ax+by+cz+d=0$ from origin may be (a,b,c,d $\in$ I) is

(C) The value(s) of λ for which the triangle with vertices A(6,10,10) B(1,0,-5) and C(6,-10, λ) will be a right angled triangle (right angled at A) is /are

(D)d is the perpendicular distance from (1, 3, 4) to $\frac{x-1}{-1}=\frac{y-1}{1}=\frac{z}{1}$, then value of $\frac{d}{2\sqrt{3}}$

Column II

(P) 0

(Q) 70/3

(R) $\sqrt{\frac{2}{3}}$

(S) $\frac{3}{2}\sqrt{65}$

65. Match the statements/expressions given in Column I with the values given in Column II

Column I

Consider a cube

(A) Angle between any two solid diagonal

(B)Angle betwen a solid diagonal and a plane

(C) Angle between plane diagonals of adjacent faces

(D)If a line makes angle $\frac{\pi}{4}$ and $\frac{\pi}{3}$ with positive X and Y axis then the angle which it makes with positive Z-axis

Column II

(P) $\cos^{-1}\frac{2}{\sqrt{6}}$

(Q) $\cos^{-1}\left(+\frac{1}{2}\right)$

(R) $\cos^{-1}\frac{1}{3}$

(S) $\frac{1}{2}$

66. Match the statements/expressions given in Column I with the values given in Column II

3x-6y+2z+10=0, Length of the perpendicular from

Column I	**Column II**
(A) origin	(P) 13/7
(B) (3, 6, 2)	(Q) 2
(C) (2, -3, 6)	(R) 10/7
(D) (–6, 2, 3)	(S) 46/7

67. Match the statements/expressions given in Column I with the values given in Column II

Column I

(A) If Acute and obtuse angle bisectors $2x-y+2z+3=0$ and $3x-2y+6z+8=0$ are represented by A and O, then

(B) If acute and obtuse angle bisectors of the planes $x-2y+2z-3=0$ and $2x-3y+6z+8=0$ are represented by A and O, then

(C) The acute and obtuse angle bisectors of the planes $2x+y-2z+3=0$ and $6x+2y-3z-8=0$ are represented by A and O, then

Column II

(P) $A : 32x+13y-23z-3=0$

(Q) $O : x-5y-4z-45=0$

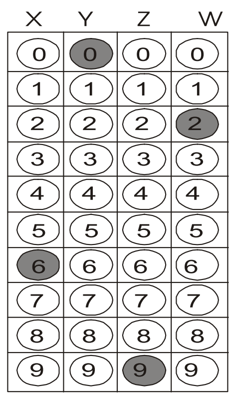
(R) $A : 23x-13y+32z+45=0$

(S) $O : 4x-y+5z-45=0$

(T) $A : 13x-23y+32z+3=0$

INTEGER QUESTIONS

The answer to each of the question is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X,Y,Z and W (say) are 6,0,9 and 2, respectively, then the correct darkening of bubbles will look like the following:



68. **If XZ plane divide the join of points (2,3,4) and (1,-1,5) in the ratio $\lambda:1$ then the integer λ should be equal to**

69. **If the area of the triangle whose vertices are A(1, 2, 3), B(2, –1, 1) and C(1, 2, –4) is λ sq unit then $\frac{2\lambda}{\sqrt{10}}$ must be**

70. **The equation of a plane which bisects the line joining (1,5,7) and (-3,1,-1) is $x+y+2z=\lambda$ then λ must be**

71. **The distance of the point $(3,0,5)$ from the line $x-2y+2z-4=0=x+3z-11$ is**

72. **The $\frac{x+4}{3}=\frac{y+6}{5}=\frac{z-1}{-2}$ and $3x-2y+z+5=0=2x+3y+4z-k$ are coplanar for k is equal to**

73. **If the distance of the point P(4, 3, 5) from the axis of y is λ unit, then the value of $\frac{5\lambda^2}{41}$ must be**

74. **Let L be the distance between the lines $x=0,\frac{y}{b}+\frac{z}{c}=1$ and $y=0,\frac{x}{a}-\frac{z}{c}=1$. Then $L^2\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)$ is**

KEY
LEVEL -V

SINGLE ANSWER QUESTIONS

1. (D)	2.(D)	3.(B)	4. (B)
5. (A)	6.(A)	7.(D)	8. (D)
9. (B)	10.(D)	11.(A)	12. (B)
13. (C)	14.(C)	15.(A)	16. (A)
17. (C)	18.(D)	19.(A,C,D)	

20. (A,B,C,D) 21. (A,B,C,D)
22.(A,C) 23.(B,C) 24. (B,C)
25. (A,B,C,D) 26.(A,B)
27.(B,D) 28. (A,C)
29. (B,D) 30.(A,C) 31.(A,B,C)
32. (A,C) 33. (B,C) 34.(B,C)
35.(B,D) 36. (A,B,C)37. (A,D)

38. (B) 39. (C) 40. (C) 41. (C)
42. (B) 43. (A) 44. (B) 45. (C)
46. (C) 47. (B) 48. (B) 49. (D)
50. (C) 51) (D) 52) (B) 53) (C)
54) (A) 55) (B) 56) (C) 57. (A)
58. (D) 59. (D) 60.(C) 61. (D)
62. (A) 63. (D)

64.(A) $\to$ (S);(B) $\to$ (P,Q,R,S);(C) $\to$ (Q);(D) $\to$ (R)

65. (A) $\to$ (R);(B) $\to$ (P);(C) $\to$ (q);(D) $\to$ (Q)

66. (A) $\to$ (R);(B) $\to$ (P);(C) $\to$ (S);(D) $\to$ (Q)

67. (A) $\to$ (R);(B) $\to$ (Q, T);(C) $\to$ (P, S)

68. (3) 69.(7) 70. (8)
71. (3) 72. (4) 73. (5) 74. (4)

HINTS
LEVEL -V

1. We have,

$(OP)^2 = 0^2 + 5^2 + 6^2 = 0 + 25 + 36 = 61$

Similarly, $(OQ)^2 = 1 + 16 = 49 = 66$

$(OR)^2 = 4 + 9 + 49 = 62$

and $(OS)^2 = 9 + 16 + 36 = 61$

So, P and S are equidistant from O and the nearst to it.

2. As $z = 1$ is common in all four points. Consider A(1,-1), B(1,3), C(4,3) and D(4,-1). ΔABC is right angled triangle with A & C as end points of diameter $\Rightarrow$ equation of circum circle of ΔABC is $(x-1)(x-4)+(y+1)(y-3)=0$
Point D(4,-1) is also satisfying it.
Hence $\square ABCD$ is cyclic quadrilateral.

3. Let the yz–plane divide the line joining the given points in the ratio $m_1 : m_2$. Then the coordinates of the point of division are

$$\left(\frac{-2m_1+3m_2}{m_1+m_2}, \frac{m_1+5m_2}{m_1+m_2}, \frac{8m_1-7m_2}{m_1+m_2}\right).$$

Since this point lies on the yz–plane, its x-coordinates is zero. Therefore
$-2m_1 + 3m_2 = 0$, i.e. $m_1 : m_2 = 3 : 2$
The other coordinates of the point of division are now

$$y = \frac{m_1+5m_2}{m_1+m_2} = \frac{3+2.5}{3+5} = \frac{13}{5}, \text{ and}$$

$$z = \frac{8m_1-7m_2}{m_1+m_2} = \frac{3.8-2.7}{3+2} = 2$$

$\Rightarrow \quad x + 5y + z = 15$

4. $(PQ)^2 = (\lambda-1)^2 + (\lambda-1)^2 + (\lambda-1)^2$

$= 3(\lambda-1)^2 = 27 \Rightarrow \quad (\lambda-1)^2 = 9$

$\Rightarrow \quad \lambda = -2 \text{ or } 4$

5. Consider the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$, therefore A is (a, 0, 0), B is (0, b, 0) and C is (0, 0, c).

Now area $\Delta ABC = \frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$

$= \frac{1}{2}|\left(-a\hat{i}+b\hat{j}\right)\times(-ai+ck)|$

$= \frac{1}{2}|(-ac)(\hat{i}\times\hat{k}) - ba(\hat{j}\times\hat{i} + bc(\hat{j}\times\hat{k})|$

$= \frac{1}{2}|(ac)\hat{j} + (ab)\hat{k} + (bc)\hat{i}|$

$= \frac{1}{2}\sqrt{(bc)^2+(ac)^2+(ab)^2}$

$= \frac{1}{2}\sqrt{1(2)^2+1^2(-1)^2+2^2(-1)^2} = \frac{3}{2}$

6. Let OP = OQ = r
M is the mid point of PQ, then coordinates of

$$M\left[\left(\frac{l_1+l_2}{2}\right)r, \left(\frac{m_1+m_2}{2}\right)r, \left(\frac{n_1+n_2}{2}\right)r\right]$$

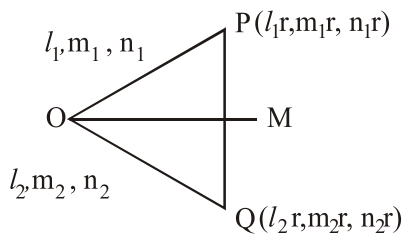
$$\left\{\left(\frac{l_1+l_2}{2}\right)r, \left(\frac{m_1+m_2}{2}\right)r, \left(\frac{n_1+n_2}{2}\right)r\right\}$$

DR's of the bisector are

$l_1 + l_2, m_1, +m_2, n_1 + n_2$

DR's of other bisector are

$l_1 - l_2, m_1 - m_2, n_1 - n_2$



7. Let the plane be $3x + y - 1 + \lambda(z-4) = 0$

It is parallel to line $\frac{x+2}{1} = \frac{y-1}{-2} = \frac{z}{1}$

$\Rightarrow \lambda = -1 \qquad \Rightarrow 3x + y - z + 3 = 0$

Hence point is $(-1, 0, 0)$

8. Line can be written as $\frac{x-1}{1} = \frac{y-0}{3} = \frac{z-1}{-1}$

If line is lying in plane the

$1(1) + 3(1) + c(-1) = 0 \quad , \qquad c = 4$

and $(1,0,1)$ is lying on plane.

$1 + 0 + c = d \Rightarrow d = 5 \quad , \quad (c+d) = 9$

9. Any point on the given line is $(t, 2t, 3t)$.

It lies in the given plane if $2(t) - 4(2t) + 2(3t) = 3$

$\Rightarrow 0 = 3$. Which is not true for any $t \in R$. Hence, the given line and given plane does not meet in any point.

10. Required plane is $2x + 2y + 2z - 11 = 0$

$\Rightarrow 2k^2 + 8k - 9 = 0 \qquad \Rightarrow k_1k_2 = -\frac{9}{2}$

11. $(x - y - z - 2) + \lambda(x + 2y + z - 2) = 0$

$(\lambda+1)x + (2\lambda-1)y + (\lambda-1)z - 2\lambda - 2 = 0$

$(\lambda+1).1 + (2\lambda-1)(-1) + (\lambda-1)(-1) = 0$

$\Rightarrow$ equation of plane is $5x + 4y + z - 10 = 0$

12. Let $A(1,-2,3)$ and $B(4,2,-1)$. Let the plane XOY meet the line AB in the point C such that C divides A and B in the ratio k : 1, then

$C \equiv \left(\frac{4k+1}{k+1}, \frac{2k-2}{k+1}, \frac{-k+3}{k+1}\right)$.

Since C lies on the plane XOY *i.e.,* the plane

$z = 0$, therefore, $\frac{-k+3}{k+1} = 0 \Rightarrow k = 3$

13. For domain of $f(x)$ we must have

$\log_{\frac{1}{2}}\left(\frac{4x-25}{x-21}\right) \geq 0 \Rightarrow 0 < \frac{4x-25}{x-21} \leq 1$

$\Rightarrow x \in \left[\frac{4}{3}, \frac{25}{4}\right)$

Integers in the domain are 2, 3 , ..., 6.

$\Rightarrow a_1 = 2, a_2 = 3, ..., a_5 = 6$

L: $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-4}{6} = r(let)$

For xy plane, put $z = 0$

$\Rightarrow 6r + 4 = 0 \Rightarrow r = \frac{-2}{3}$

$A(2r+1, 3r-2, 0) = \left(-\frac{1}{3}, -4, 0\right)$

At yz plane, put $x = 0$

$\Rightarrow 2r + 1 = 0 \quad \Rightarrow r = \frac{-1}{2}$

$B(0, 3r-2, 6r+4) = \left(0, \frac{-7}{2}, 1\right)$

At zx plane, put $y = 0 \Rightarrow 3r - 2 = 0 \Rightarrow r = \frac{2}{3}$

$C(2r+1, 0, 6r+4) = \left(\frac{7}{3}, 0, 8\right) \Rightarrow C'\left(\frac{7}{3}, 0, -8\right)$

Volume of the tetrahedron

$OABC' = V = \frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}]$

$$V = \frac{1}{6}\begin{vmatrix} -1/3 & -4 & 0 \\ 0 & -7/2 & 1 \\ 7/3 & 0 & -8 \end{vmatrix} = \left|-\frac{28}{9}\right|$$

Hence $\quad 90V = 280$

MULTI ANSWER QUESTIONS

14. Given $|\vec{b} - \vec{a}| = 12$; $|\vec{c}| = 6$

Equation of CD is $\vec{r} = \lambda\vec{c}$

& eq. of AB is $\vec{r} = \vec{a} + \mu(\vec{b} - \vec{a})$

$$S.D. = \left|\frac{\vec{a}\cdot\vec{c}\times(\vec{b}-\vec{a})}{\vec{c}\times(\vec{b}-\vec{a})}\right| = 8$$

$$\left|\frac{[\vec{a}\ \vec{c}\ \vec{b}]}{\vec{c}\times(\vec{b}-\vec{a})}\right| = 8, \ \left|[\vec{a}\ \vec{c}\ \vec{b}]\right| = 8\left|\vec{c}\times(\vec{b}-\vec{a})\right|$$

15. Given equation of straight line

$\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$

Since, the line lies in the plane $2x - 4y + z = 7$

$\therefore$ point (4,2,k) must satisfy the plane

$\Rightarrow 8 - 8 + k = 7 \Rightarrow k = 7$

16. Given $\overrightarrow{OQ} = (1-3\mu)\hat{i} + (\mu-1)\hat{j} + (5\mu+2)\hat{k}$

$\overrightarrow{OP} = 3\hat{i} + 2\hat{j} + 6\hat{k}$ (where O is origin)

Now $\overrightarrow{PQ} = (1-3\mu-3)\hat{i} + (\mu-1-2)\hat{j} + (5\mu+2-6)\hat{k}$

$= (-2-3\mu)\hat{i} + (\mu-3)\hat{j} + (5\mu-4)\hat{k}$

$\because \overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$

$\therefore -2 - 3\mu - 4\mu + 12 + 15\mu - 12 = 0$

$\Rightarrow 8\mu = 2 \Rightarrow \mu = \frac{1}{4}$

17. Since $\ell = m = n = \frac{1}{\sqrt{3}}$

$\therefore$ Equation of line are $\frac{x-2}{1/\sqrt{3}}=\frac{y+1}{1/\sqrt{3}}=\frac{z-2}{1/\sqrt{3}}$

$\Rightarrow x-2=y+1=z-2=r$ (say)

$\therefore$ Any point on the line is

$Q\equiv(r+2,r-1,r+2)$

$\because$ Q lies on the plane $2x+y=z=9$

$\therefore\ 2(r+2)+(r-1)+(r+2)=9$

$\Rightarrow 4r+5=9\Rightarrow r=1$

$\therefore\ PQ=\sqrt{(3-2)^2+(0+1)^2+(3-2)^2}=\sqrt{3}$

18. Let the equation of plane be

$a(x-1)+b(y+2)+c(z-1)=0$

Which is perpendicular to $2x-2y+z=0$ and $x-y+\ 2z=4$

$\Rightarrow 2a-2b+c=0$ and $a-b+2c=0$

$\Rightarrow\frac{a}{-3}=\frac{b}{-3}=\frac{c}{0}\Rightarrow\frac{a}{1}=\frac{b}{1}=\frac{c}{0}$

So, the equation of plane is

$x-1+y+2=0$ or $x+y+1=0$

Its distance from the point $(1,2,2)$ is

$\frac{|1+2+1|}{\sqrt{2}}=2\sqrt{2}$

19. $\because\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1\quad\ldots$(i)

$\Rightarrow 1-\sin^2\alpha+1-\sin^2\beta+\cos^2\gamma=1$

or $\sin^2\alpha+\sin^2\beta=1+\cos^2\gamma$

also from (i),

$\cos^2\beta+\cos^2\gamma=1-\cos^2\alpha$

or $\cos^2\beta+\cos^2\gamma=\sin^2\alpha$

$\Rightarrow\sin^2\alpha=\cos^2\beta+\cos^2\gamma$

20. Let the line be inclined at an angle α with each of the three coordinate axes, that the direction cosines of the line are $\cos\alpha$, $\cos\alpha$, $\cos\alpha$ and

$\cos^2\alpha+\cos^2\alpha+\cos^2\alpha=1\Rightarrow$

$\cos^2\alpha=1/3\Rightarrow\cos\alpha=\pm1/\sqrt{3}$

21. $l+m+n=0$ and $2mn+2ml-nl=0$

$l=-(m+n)\Rightarrow 2mn-(2m-n)(m+n)=0$

$\Rightarrow 2m^2-n^2-mn=0\Rightarrow\frac{m}{n}=1\ or\ \frac{-1}{2}$

when $m=n$, $l=-2n$; when $m=\frac{-n}{2},l=\frac{-n}{2}$

$hence=\frac{l}{-2}=\frac{m}{1}=\frac{n}{1}\ or\ \frac{l}{-1}=\frac{m}{-1}=\frac{n}{2}$

Since $l^2+m^2+n^2=1$

$(l,m,n)=\left(\frac{-2}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}}\right)or\left(\frac{-1}{\sqrt{6}},\frac{-1}{\sqrt{6}},\frac{2}{\sqrt{6}}\right)$

$or\left(\frac{2}{\sqrt{6}},\frac{-1}{\sqrt{6}},\frac{-1}{\sqrt{6}}\right)or\left(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{-2}{\sqrt{6}}\right)$

22. The given lines are coplanar, if

$$0=\begin{vmatrix}2-1 & 3-4 & 4-5\\ 1 & 1 & -k\\ k & 2 & 1\end{vmatrix}=\begin{vmatrix}1 & -1 & -1\\ 1 & 1 & -k\\ k & 2 & 1\end{vmatrix}$$

or $2(1+k)-(k+2)(1-k)=0$

or $k^2+3k=0\ \Rightarrow k=0,-3$

23. For the given lines

$$\begin{vmatrix}4-1 & 0-1 & -1-(-1)\\ 3 & -1 & 0\\ 2 & 0 & 3\end{vmatrix}=0$$

So, the given lines intersect.

Any point on the first line is $(3r_1+1,-r_1+1,-1)$ and any point on the second line is $(2r_2+4,0,3r_2-1)$.

Since, the lines intersect, at the point of intersection.

$3r_1+1=2r_2+4,-r_1+1=0,-1=3r_2-1$

$r_1=1,r_2=0$

Hence, the point of intersection is (4, 0, –1)

24. d.r. of given line are (1,–2,3) and the direction ratios of normal to plane are (1, 2, 1).

Since $1\times1+(-2)\times2+3\times1=0,$ therefore the line is parallel to the plane.

Also, the base point of the line (1, 2, 1) lies in the given plane $(1+2\times2+1=6$ is true)

Hence, the given line lies in the given plane.

25. (a) We know that the plane $ax+by+cz+d=0$

contains the line $\frac{x-\alpha}{l}=\frac{y-\beta}{m}=\frac{z-\gamma}{n}$

if $a\alpha+b\beta+c\gamma+d=0$ and $al+bm+cn=0.$

Now, since (–1) – 2 (3) + 7 (–2) + 21 = 0
And (–3) (1) + 2 (–2) + 1 (7) = 0
The line given in (a) lies on the given plane.
(b) Since, $0-2(7)+7(-1)+21=0$
The point (0, 7, –1) lies on the plane.
(c) Direction ratio of the normal to the given plane are (1, -2, 7) which are same as those of the given in (c). So, the plane is perpendicular to the lines.
(d) direction ratios of normal to plane are equal hence two planes are parallel.

26. The given equation are
$4x-4y-z+11=0 \quad \ldots(i)$
$x+2y-z-1=0 \quad \ldots(ii)$
The D.r's of normals to the planes (i) and (ii) are 4, –4, –1 and 1, 2, –1 respectively.
Let Dr's of line of intersection of plane be l, m, n
As the line of intersection of the planes isperpendicular to the normals of the both planes, we get
$$4l-4m-n=0$$
and $$l+2m-n=0$$
By cross multiplication $\frac{l}{6}=\frac{m}{3}=\frac{n}{12}$ or $\frac{l}{2}=\frac{m}{1}=\frac{n}{4}$
If x = 0, Eqs. (i) and (ii) becomes
$$-4y-z+11=0$$
$$2y-z-1=0$$
Solving, we get $y=2, z=3$
Equation of line is $\frac{x}{2}=\frac{y-2}{1}=\frac{z-3}{4}$
Also x = 4, y = 4, z = 11 satisfies Eqs. (i) and (ii)
Hence, (b) is also the correct option.

27. $\because \alpha=60°, \beta=60°$
$$\therefore \cos\alpha=\cos\beta=\frac{1}{2}$$
$$\Rightarrow \cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$$
$$\Rightarrow \frac{1}{4}+\frac{1}{4}+\cos^2\gamma=1 \Rightarrow \cos^2\gamma=\frac{1}{2}$$
$$\therefore \cos\gamma=\pm\frac{1}{\sqrt{2}} \Rightarrow \gamma=\frac{\pi}{4},\frac{3\pi}{4}$$

28. The given lines (in symmetrical form) are
$$\frac{x-1}{3}=\frac{y-1}{-1}=\frac{z-(-1)}{0}$$
$$\frac{x-4}{2}=\frac{y-0}{0}=\frac{z-(-1)}{3}$$
Clearly, the lines are not parallel as their d.r.s. are not proportinal. These lines are coplanar
$$\text{iff} \begin{vmatrix} 4-1 & 0-1 & -1-(-1) \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix}=0,$$
$$\text{i.e. iff} \begin{vmatrix} 3 & -1 & 0 \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix}=0,$$
which is true. Hence the lines are coplanar; being non-parallel, they meet in an unique point.

29. Dr's of normal to the plane are (2, -3, 6)
Dr's of x-axis are (1, 0, 0)
If θ is angle between the plane and x-axis then $\frac{\pi}{2}-\theta$ is the angle between the x-axis and normal to the plane.
$$\therefore \cos\left(\frac{\pi}{2}-\theta\right)=\frac{2}{\sqrt{2^2+(-3)^2+6^2}}=\frac{2}{7}$$
$$\sin\theta=\frac{2}{7} \ \& \ \tan\theta=\frac{2}{3\sqrt{5}}$$

30. If θ is the angle between the line $\vec{r}=\vec{a}+\lambda\vec{b}$ and the plane $\vec{r}.\vec{n}=p$ then $90^0-\theta$ is the angle between this line and the normal vector $\hat{n}$ to the plane
$$\cos(90^0-\theta)=\frac{\vec{b}.\hat{n}}{|\vec{b}|}$$
$$\theta=\frac{\pi}{2}-\cos^{-1}\left(\frac{\vec{b}.\hat{n}}{|\vec{b}|}\right)$$
$$\sin\theta=\frac{\vec{b}.\hat{n}}{|\vec{b}|} \Rightarrow \theta=\sin^{-1}\left(\frac{\vec{b}.\hat{n}}{|\vec{b}|}\right)$$

31. At $(0,0,0), x-y-z-2=-2=(-ve)$
at (2, 3, 1), $x-y-z-2=2-3-1-2=-4$
Since, both have same sign (0, 0, 0) and (2, 3, 1) lie on the same side of the plane.

Distance $= \left|\dfrac{2-3-1-2}{\sqrt{1^2+1^2+1^2}}\right| = \dfrac{4}{\sqrt{3}}$

Equation of a line perpendicular to the plane $x-y-z-2=$ and passing through the point (2, 3, 1) is

$\dfrac{x-2}{1}=\dfrac{y-3}{-1}=\dfrac{z-1}{-1}=\lambda$

A point on the line is $(\lambda+2, 3-\lambda, 1-\lambda)$ and it lies and the plane $x-y-z-2=0$

if $\lambda+2-3+\lambda-1=\lambda-2=0 \Rightarrow \quad \lambda=\dfrac{4}{3}$

Foot of perpendicular on the plane is

$\left(\dfrac{4}{3}+2, 3-\dfrac{4}{3}, 1-\dfrac{4}{3}\right)=\left(\dfrac{10}{3}, \dfrac{5}{3}, -\dfrac{1}{3}\right)$

32. The equation of the plane through $(2,3,-1)$ and perpendicular to the vector $3\hat{i}-4\hat{j}+7\hat{k}$ is

$3(x-2)+(-4)(y-3)+7(z-(-1))=0$

or $3x-4y+7z+13=0$

Distance of this plane from the origin

$=\dfrac{|3\times 0-4\times 0+7\times 0+13|}{\sqrt{3^2+(-4)^2+7^2}}=\dfrac{13}{\sqrt{74}}.$

33. The given plane is $2x-y+z+3=0$

Note that the point (1, 3, 4) does not lie in plane. So, the projection of the point in the plane is the foot of perpendicular from (1,3 4) onto the given plane. The equations of the line through (1, 3, 4) and at right angles to the given plane are

$\dfrac{x-1}{2}=\dfrac{y-3}{-1}=\dfrac{z-4}{1}$ ...(1)

Any pont on this line is $(2t+1, -t+3, t+4)$

It lies in the given plane

if $2(2t+1)-(-t+3)+(t+4)+3=0$

i.e., if $6t+6=0$ i.e., if $t=-1$.

So, the required projection is

$(2\times(-1)+1-(-1)+3, -1+4)$

i.e., (–1, 4, 3)

34. Let the plane be

$\left|\dfrac{-1}{\sqrt{\left(\dfrac{1}{a}\right)^2+\left(\dfrac{1}{b}\right)^2+\left(\dfrac{1}{c}\right)^2}}\right|=1$ or $\dfrac{1}{a^2}+\dfrac{1}{b^2}+\dfrac{1}{c^2}=1$

The plane cuts the coordinate axes at A(a, 0, 0) B(0,b,0), C(0,0,c). The centroid of $\Delta\, ABC$ is

$\left(\dfrac{a}{3}, \dfrac{b}{3}, \dfrac{c}{3}\right)=(x,y,z)$

(let) $x^{-2}+y^{-2}+z^{-2}=9$

35. Equation of the line passing through P(1, 4, 3) is

$\dfrac{x-1}{a}=\dfrac{y-4}{b}=\dfrac{z-3}{c}$ ....(1)

Since (1) is perpendicular to

$\dfrac{x-1}{2}=\dfrac{y+3}{1}=\dfrac{z-2}{4}$ and

$\dfrac{x+2}{3}=\dfrac{y-4}{2}=\dfrac{z+1}{-2}$

Hence $\quad 2a+b+4c=0$

and $\quad 3a+2b-2c=0$

$\dfrac{a}{-2-8}=\dfrac{b}{12+4}=\dfrac{c}{4-3}$

$\Rightarrow \dfrac{a}{-10}=\dfrac{b}{16}=\dfrac{c}{1}$

Hence the equation of the lines is

$\dfrac{x-1}{-10}=\dfrac{y-4}{16}=\dfrac{z-3}{1}$ **....(2)**

Ans. Now any point Q on (2) can be taken as

$(1-10l, 16l+4, l+3)$

Distance of Q from P (1, 4, 3)

$=(10l)^2+(16l)^2+l^2=357$

$\Rightarrow (100+256+1)l^2=357 \Rightarrow l=1$ or -1

Q is (–9, 20, 4) or (11, – 12, 2)

Hence $\quad a_1+a_2+a_3=15$ or 1

36. Let $\overrightarrow{OA}=\vec{a}, \overrightarrow{OB}=\vec{b}, \overrightarrow{OC}=\vec{c}$, then

$\vec{a}.\vec{a}+(\vec{b}-\vec{c}).(\vec{b}-\vec{c})=\vec{b}.\vec{b}+(\vec{c}-\vec{a}).(\vec{c}-\vec{a})$

$\Rightarrow -2\vec{b}.\vec{c}=-2\vec{c}.\vec{a} \Rightarrow (\vec{a}-\vec{b}).\vec{c}=0$ or $\overrightarrow{BA}.\overrightarrow{OC}=0$

Hence, $AB \perp OC$

Similarly, $BC \perp OA$ and $CA \perp OB$

37. Take OB as X-axis. we have $B=\left(4\sqrt{2},0,0\right)$,

$A=\left(0,0,2\right)$ *and* $C=\left(2\sqrt{2},2\sqrt{6},0\right)$.

Mid point of OB is $\left(2\sqrt{2},0,0\right)$, Mid point of BC is $\left(3\sqrt{2},\sqrt{6},0\right)$, hence the direction ratios of the lines be $2\sqrt{2}:0:-2$ and $3\sqrt{2}:\sqrt{10}:0$

COMPREHENSION QUESTIONS

38. Consider line passing through $P(1,-2,3)$

$$\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-4}=\lambda$$

Q on line $Q(2\lambda+1,3\lambda-2,-4\lambda+3)$ is also lying on plane.

$$(2\lambda+1)-(3\lambda-2)+(-4\lambda+3)=5\Rightarrow-5\lambda=-1\Rightarrow\lambda=\frac{1}{5}$$

$$PQ=\sqrt{(2\lambda)^2+(3\lambda)^2+(4\lambda)^2}=\frac{\sqrt{29}}{5}$$

39. Normal to plane.

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0-(-3) & z-1 & 4-2\\ 3 & 4 & -2\end{vmatrix}=-10\hat{i}+12\hat{j}+9\hat{k}$$

Equation of plane is

$$-10(x-0)+12(y-2)+9(z-4)=0$$

$$-10x+12y-24+9z-36=0$$

$$10x-12y-9z+60=0$$

40. Let the position vector of L be

$$\vec{a}+\lambda\vec{b}=(6+3\lambda)\hat{i}+(7+2\lambda)\hat{j}+(7-2\lambda)\hat{k}$$

So $\overrightarrow{PL}=(6+3\lambda)\hat{i}+(7+2\lambda)\hat{j}+(7-2\lambda)\hat{k}-\left(\hat{i}+2\hat{j}+3\hat{k}\right)$

$$=(5+3\lambda)\hat{i}+(5+2\lambda)\hat{j}+(4-2\lambda)\hat{k}$$

Since $\overrightarrow{PL}$ is perpendicular to the given line which is parallel to $\vec{b}=3\hat{i}+2\hat{j}-2\hat{k}$

$$\Rightarrow 3(5+3\lambda)+2(5+2\lambda)-2(4-2\lambda)=0$$

$\Rightarrow \lambda=-1$ and thus the position vector of L is $3\hat{i}+5\hat{j}+9\hat{k}$

41. Let the position vector of Q, the image of P in the given line be $x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$, then L is the mid-point of PQ.

$$\Rightarrow 3\hat{i}+5\hat{j}+9\hat{k}=\frac{\hat{i}+2\hat{j}+3\hat{k}+x_1\hat{i}+y_1\hat{j}+z_1\hat{k}}{2}$$

$$\Rightarrow \frac{x_1+1}{2}=3,\ \frac{y_1+2}{2}=5,\ \frac{z_1+3}{2}=9$$

$$\Rightarrow x_1=5,\ y_1=8, z_1=15$$

$\Rightarrow$ image of P in the line is (5, 8, 15)

42. Area of the

$$\Delta PLA=\frac{1}{2}\left|\overline{PL}\right|\left|\overline{AL}\right|=\frac{1}{2}\left|2\hat{i}+3\hat{j}+6\hat{k}\right|\left|-3\hat{i}-2\hat{j}+2\hat{k}\right|$$

$$=\frac{1}{2}\sqrt{4+9+36}\ \sqrt{9+4+4}=\frac{7\sqrt{17}}{2}\text{ sq. uints.}$$

43. Let P(x, y, z) be any point on the locus then 3PA $=2PB\Rightarrow 9(PA)^2=4(PB)^2$

$$\Rightarrow 9\left[(x+2)^2+(y-2)^2+(z-3)^2\right]=4$$

$$\left[(x-13)^2+(y+3)^2+(z-13)^2\right]$$

$$\Rightarrow 5\left(x^2+y^2+z^2\right)+140x-60y+50z-1235=0$$

$$\Rightarrow x^2+y^2+z^2+28x-12y+10z-247=0$$

44. The required coordinates are

$$\left(\frac{2\times13+3(-2)}{2+3},\frac{2\times(-3)+3(2)}{2+3},\frac{2\times13+3\times3}{2+3}\right)$$

= (4, 0, 7)

45. Direction ratios of AB are 13+2, -3-2, 13-3 i.e. 15, -5, 10

Let the equation of the required line L be

$$\frac{x+2}{l}=\frac{y-2}{m}=\frac{z-3}{n}$$

then $15l-5m+10n=0$

which satisfied by (c)

46. The equation of plane is $3x-y+2z-7=0$.

Let the line segment AB cuts the plane in the ratio

$\mu:1\ \Rightarrow C\left(\frac{1}{\mu+1},\frac{6\mu-4}{\mu+1},\frac{\mu+5}{\mu+1}\right)$

on $3x-y+2z=7$

$$\therefore 3\left(\frac{1}{\mu+1}\right)-\left(\frac{6\mu-4}{\mu+1}\right)+2\left(\frac{\mu+5}{\mu+1}\right)=7$$

$$\Rightarrow \mu=\frac{10}{11}$$

47. As A and P are on same side of the plane, the value of $3x-y+2z-7$ has same sign at A and

P. $\Rightarrow 3\left(\lambda^2+1\right)-\lambda+2\left(\lambda-1\right)-7>0$

or $3\lambda^2+\lambda-6>0$

$$\therefore \left(\lambda-\frac{-1-\sqrt{73}}{6}\right)\left(\lambda-\frac{\sqrt{73}-1}{6}\right)>0$$

$$\Rightarrow \lambda \in \left(-\infty, \frac{-1-\sqrt{73}}{6}\right) \cup \left(\frac{-1+\sqrt{73}}{6}, \infty\right)$$

48. The equations of given lines in vector forms may be written as

$$L_1 : \vec{r} = \left(-\hat{i}-2\hat{j}-\hat{k}\right)+\lambda\left(3\hat{i}+\hat{j}+2\hat{k}\right)$$

and $L_2 : \vec{r} = \left(2\hat{i}-2\hat{j}+3\hat{k}\right)+\mu\left(\hat{i}+2\hat{j}+3\hat{k}\right)$

Since, the vector perpendicular to both L_1 and L_2

$$\therefore \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = -\hat{i}-7\hat{j}+5\hat{k}$$

$\therefore$ required unit vector

$$= \frac{\left(-\hat{i}-7\hat{j}+5\hat{k}\right)}{\sqrt{(-1)^2+(-7)^2+(-5)^2}} = \frac{1}{5\sqrt{3}}\left(-\hat{i}-7\hat{j}+5\hat{k}\right)$$

49. The shortest distance between L_1 and L_2 is

$$\left|\frac{\left\{\left(2-(-1)\right)\hat{i}+\left(2-2\right)\hat{j}+\left(3-(-1)\right)\hat{k}\right\}.\left(-\hat{i}-7\hat{j}+5\hat{k}\right)}{5\sqrt{3}}\right|$$

$$= \left|\frac{\left(3\hat{i}+4\hat{k}\right).\left(-\hat{i}-7\hat{j}+5\hat{k}\right)}{5\sqrt{3}}\right| = \frac{17}{5\sqrt{3}} \text{unit}$$

50. The equation of the plane passing through the point $\left(-1,-2,-1\right)$ and whose normal is prependiuclr to the both the given lines L_1 and L_2 may be written as

$$\left(x+1\right)+7\left(y+2\right)-5\left(z+1\right)=0$$

$$\Rightarrow x+7y-5z+10=0$$

The distance of the point (1,1,1) from the plane

$$= \left|\frac{1+7-5+10}{\sqrt{1+49+25}}\right| = \frac{13}{\sqrt{75}} \text{ unit.}$$

PASSAGE:

To find the equation of L_1 and L_2 in symmetrical form:

$L_1 : x = 0,\ cy + bz - bc = 0$

If d·r's of L_1 be a_1, b_1, c_1 then $1 \cdot a_1 + 0 \cdot b_1 + 0 \cdot c_1 = 0$ and $0 \cdot a_1 + c \cdot b_1 + b \cdot c_1 = 0$, so

$\frac{a_1}{0} = \frac{b_1}{-b} = \frac{c_1}{c} \Rightarrow$ d·r's of L_1 will be $0, -b, c$.

For any point on L_1, let $z = 0$, so the point will be (0, b, 0)

Equation of L_1 in symmetrical form will be

$$\frac{x}{0} = \frac{y-b}{-b} = \frac{z}{c}$$

Similarly, equation of L_2 in symmetrical form will be $\frac{x-a}{a} = \frac{y}{0} = \frac{z}{c}$

51. Equation of plane 'P' will be

$$\begin{vmatrix} x & y-b & z \\ 0 & -b & c \\ a & 0 & c \end{vmatrix} = 0$$

$$\Rightarrow -bc \cdot x + ac\,(y-b) + ab \cdot z = 0$$

$$\Rightarrow \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$$

$\Rightarrow$ Option (D) is correct.

52. Vector equation of L_1 and L_2 is

$L_1 : \vec{r} = \vec{p}_1 + \lambda\vec{q}_1 \Rightarrow \vec{r} = b\hat{j} + \lambda(-b\hat{j} + c\hat{k})$ and

$L_2 : \vec{r} = \vec{p}_2 + \mu\vec{q}_2 \Rightarrow \vec{r} = a\hat{i} + \mu(a\hat{i} + c\hat{k})$

Now shortest distance between L_1 and L_2

$$= \frac{[\vec{p}_1 - \vec{p}_2\ \vec{q}_1\ \vec{q}_2]}{|\vec{q}_1 \times \vec{q}_2|} = \frac{\begin{vmatrix} -a & b & 0 \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}}{\sqrt{(bc)^2+(ac)^2+(ab)^2}} = \frac{1}{4}$$

(given)

$$\Rightarrow \left|\frac{2abc}{\sqrt{(bc)^2+(ac)^2+(ab)^2}}\right| = \frac{1}{4}$$

$$\Rightarrow \frac{(bc)^2+(ac)^2+(ab)^2}{a^2b^2c^2} = 64$$

Hence $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 64$

$\Rightarrow$ Option (B) is correct.

53. Given : $a=b=c=1$ \ normal line of the plane P from A(1, 0, 0) is $\frac{x-1}{1} = \frac{y}{-1} = \frac{z}{-1}$

$\Rightarrow \quad \frac{x-1}{-1}=\frac{y}{1}=\frac{z}{1}=1\,(\text{say})$

Any point B on the line will be $B(1-l,\ l,\ l)$

If this point be image of A in plane P then mid point C of AB lane P

$$\frac{1-\frac{\lambda}{2}}{1}-\frac{\frac{\lambda}{2}}{1}-\frac{\frac{\lambda}{2}}{1}+1=0 \Rightarrow \quad l=\frac{4}{3}$$

$$B\left(\frac{-1}{3},\frac{4}{3},\frac{4}{3}\right)$$

Hence BM $=\sqrt{\frac{81}{9}}=3$

$\Rightarrow$ Option (C) is correct.

54,55,56

We have $\frac{x-1}{2}=\frac{y}{1}=\frac{z+1}{-2}=t\ (\text{say})$

Now $\quad \overline{AP}=2t\hat{i}+(t-1)\hat{j}-2(t+1)\hat{k}$

As $\overline{AP}\cdot\vec{V}=0 \Rightarrow t=\frac{-1}{3}$

Again $a_1+1=\frac{2}{3} \Rightarrow a_1=\frac{-1}{3}$

$a_2+1=\frac{-2}{3}\Rightarrow a_2=\frac{-5}{3}$

$a_3+1=\frac{-2}{3}\Rightarrow a_3=\frac{-5}{3}$

Hence Q is $\left(\frac{-1}{3},\frac{-5}{3},\frac{-5}{3}\right)$

Hence OQ $=\sqrt{\frac{1}{9}+\frac{25}{9}+\frac{25}{9}}=\sqrt{\frac{17}{3}}$

Ans.(iii)

Equation of the plane containing the point A and L is given by $[\overrightarrow{PA},\ \overrightarrow{RA}\ \ \vec{V}]=0$

$$\Rightarrow \begin{vmatrix} x-1 & y-1 & z-1 \\ 0 & 1 & 2 \\ 2 & 1 & -2 \end{vmatrix}=0$$

$\Rightarrow (x-1)(x-2)+2\big(2(y-1)-(z-1)\big)=0$

$\Rightarrow -4(x-1)+4(y-1)-2(z-1)=0$

$\Rightarrow 2(x-1)-2(y-1)+(z-1)=0$

$\Rightarrow 2x-2y+z=1 \qquad \ldots\ldots(1)$

Distance of origin from (1) is $\frac{1}{\sqrt{9}}=\frac{1}{3}$ **Ans.(i)**

Finally AP $=\sqrt{\frac{4}{9}+\frac{16}{9}+\frac{16}{9}}=\sqrt{4}=2$ **Ans.(ii)**

ASSERTION & REASON QUESTIONS

57. Conceptual

58. If the points are collinear then it will not form a unique plane.

59. $\sin\theta=\left|\frac{2-3+2}{\sqrt{4+9+4}\sqrt{3}}\right|=\frac{1}{\sqrt{51}}$

60. is the correct option because D.R's of normal to the plane are (0, 1, 1) which implies that normal to the plane is also normal to x-axis. Hence plane and x-axis are parallel.

61. $AB=3\sqrt{2},\ BC=3\sqrt{2},\ CD=3\sqrt{2}, DA=3\sqrt{2}$

$AC=\sqrt{\left\{(2+2)^2+(9-11)^2+(12-8)^2\right\}}$

$=\sqrt{36}=6$

and $BD=\sqrt{\left\{(1+1)^2+(8-12)^2+(8-12)^2\right\}}$

$=\sqrt{36}=6$

Hence, AB = BC = CD = DA and AC=BD.

Then, it is a square not a rhombus.

62. Algebraic distance of (2, 1, 5) from the plane $2x+2y-2z-1=0$ is

$$\frac{2(2)+2(1)-2(5)-1}{\sqrt{(2)^2(2)^2+(-2)^2}}=-\frac{5}{2\sqrt{3}} \qquad \ldots.(i)$$

and algebraic distance of (3, 4, 3) from the plane $2x+2y-2z-1=0$ is

$$\frac{2(3)+2(4)-2(3)-1}{\sqrt{\left\{(2)^2+(2)^2+(-2)^2\right\}}}=\frac{7}{2\sqrt{3}} \qquad \ldots.(ii)$$

Since Eqs. (i) and (ii) are of opposite sign.

Hence, points (2, 1, 5) and (3, 4, 3) lie on opposite side of the plane 2x+2y–2z–1=0

63. Given planes are $3x-6y-2z=15$ and $2x+y-2z=5$, For $z=0$, we get $x=3,\ y=-1$

Direction ratios of planes are $(3,-6,-2)$ and $(2,1,-2)$, Then the DR's of line of intersection of planes is $(14,2,15)$ and line is

$$\frac{x-3}{14}=\frac{y+1}{2}=\frac{z-0}{15}=\lambda \qquad \text{(say)}$$

$\Rightarrow x=14\lambda+3, y=2\lambda-1, z=15\lambda$

Hence, statement 1 is false

But statement 2 is true.

MATRIX-MATCHING

64. Let O(0,0,0), A(3,4,7) and B(5,2,6) be the given point

Area of $\Delta OAB=\frac{1}{2}OA.OB\ \sin(\angle AOB)$

Now, $OA=\sqrt{3^2+4^2+7^2}=\sqrt{74}$

$OB=\sqrt{5^2+2^2+6^2}=\sqrt{65}$

Also D.c's of the line OA and OB are

$=\frac{3}{\sqrt{74}},\frac{4}{\sqrt{74}},\frac{7}{\sqrt{74}}$ and $\frac{5}{\sqrt{65}},\frac{2}{\sqrt{65}},\frac{6}{\sqrt{65}}$

$\therefore$ Required area

$\frac{1}{2}\times\sqrt{74}\times\sqrt{65}\times\frac{3}{\sqrt{74}}=\frac{3}{2}\sqrt{65}$

(B) Distance $=\frac{|d|}{\sqrt{a^2+b^2+c^2}}$

(C) Let the given points be A, B and C respectively. Then find AB, AC, BC and then apply $AB^2+AC^2=BC^2$ then solve for the λ.

(D)Any point on the line is $(1-r, r+1, r)$

The direction ratio of the line joining (1, 3, 4) & (1 –r, r + 1, r) is –r, r–2, r–4

$\therefore(-1)(-r)+1.(r-2)+(r-4)=0$

$r+r-2+r-4=0, \qquad 3r=6\Rightarrow r=2$

$\therefore$ Foot of the perpendicular is (–1, 3, +2)

$\therefore$ distance $\sqrt{(2)^2+0+4}=2\sqrt{2} \qquad \therefore d=2\sqrt{2}$

$\frac{d}{2\sqrt{3}}=\frac{2\sqrt{2}}{2\sqrt{3}}=\frac{\sqrt{2}}{\sqrt{3}}$

65. The solid diagonals may be taken as the lines joing (0, 0, 0), (a, a, a) and (a, a, 0) and (0, 0, a). The direction ratios wil be a, a, a; a, a, –a.

$\Rightarrow \cos\theta=\frac{a^2+a^2-a^2}{\sqrt{3a^2}\times\sqrt{3a^2}}\frac{1}{3}\Rightarrow \theta=\cos^{-1}\frac{1}{3}$

Let us take the solid diagonal as the one joining (0, 0, 0), (a, a, a) and plane diagonal as joining (0, 0, 0) and (a, a, 0). We easily get the angle as $\cos^{-1}\frac{2}{\sqrt{6}}$.

The third part is easily found as $\cos^{-1}\left(\frac{1}{2}\right)$

(D) $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

$\cos^2\frac{\pi}{4}+\cos^2\frac{\pi}{3}+\cos^2\gamma=1$, $\cos\gamma=\pm\frac{1}{2}$

$\gamma=\cos^{-1}\left(\frac{1}{2}\right)$

66. (A) $\frac{10}{\sqrt{9+36+4}}=\frac{10}{7}$

(B) $\frac{|3\times3-6\times6+2\times2+10|}{7}=\frac{13}{7}$

(C) $\frac{|3\times2-6(-3)+2\times6+10|}{7}=\frac{46}{7}$

(D) $\frac{|3(-6)-6(2)+2(3)+10|}{7}=\frac{14}{7}=2$

67. (A) $\because (2)(3)+(-1)(-2)+(2)(6)=20>0$

Bisectors are

$$\frac{(2x-y+2z+3)}{\sqrt{(2)^2+(-1)^2+(2)^2}}=\pm\frac{(3x-2y+6z+8)}{\sqrt{(3)^2+(-2)^2+(6)^2}}$$

or 7 (2x–y+2z+3) = ± 3(3x–2y+6z+8)

Acute angle bisector is

$7(2x-y+2z+3)=-3(3x-2y+6z+8)$

$\Rightarrow 23x-13y+32z+45=0$

and obtuse angle bisector is

$7(2x-y+2z+3)=3(3x-2y+6z+8)$

$\Rightarrow\ 5x-y-4z-3=0$

A : $23x-13y+32z+45=0$

and O : $5x–y–4z–3=0$

(B) The given planes can be written as

$-x+2y-2z+3=0$ and $2x-3y+6z+8=0$

$\because (-1)(2)+(2)(-3)+(-2)(6)$

$=-2-6-12=-20<0$

Bisectors are,

$$\frac{(-x+2y-2z+3)}{\sqrt{(-1)^2+(2)^2+(-2)^2}}=\pm\frac{(2x-3y+6z+8)}{\sqrt{(2)^2+(-3)^2+(6)^2}}$$

$\Rightarrow 7(-x+2y-2z+3)=\pm 3(2x-3y+6z+8)$

Acute angle bisector is

$7(-x+2y-2z+3)=3(2x-3y+6z+8)$

$\Rightarrow 13x-23y+32z+3=0$

and obtuse bisector is

$7(-x+2y-2z+3)=-3(2x-3y+6z+8)$

$\Rightarrow\quad x-5y-4z-45=0$

$\Rightarrow\ $ A : $13x-23y+32z+3=0$

and O : $x-5y-4z-45=0$

(C) The given planes can be written as

$2x+y-2z+3=0$ and $-6x-2y+3z+8=0$

$\because (2)(-6)+(1)(-2)+(-2)(3)=-20<0$

Bisectors are

$$\frac{(2x+y-2z+3)}{\sqrt{\{(2)^2+(1)^2+(-2)^2\}}}=\pm\frac{(-6x-2y+3z+8)}{\sqrt{\{(-6)^2+(-2)^2+(3)^2\}}}$$

$\Rightarrow 7(2x+y-2z+3)=\pm 3(-6x-2y+3z+8)$

Acute angle bisector is

$7(2x+y-2z+3)=3(-6x-2y+3z+8)$

$\Rightarrow 32x+13y-23z-3=0$

and obtuse bisector is

$7(2x+y-2z+3)=-3(-6x-2y+3z+8)$

$\Rightarrow 4x-y+5z-45=0$

A:$32x+13y–23z–3=0$

and O:$4x–y+5z–45=0$

INTEGER QUESTIONS

68. We must have $\dfrac{\lambda(-1)+1\times 3}{\lambda+1}=0\Rightarrow\lambda=3$

69. The coordinates of the projections of A, B, C on the yz-plane are (0, 2, 3), (0, -1, 1) and (0, 2, -4) respectively

$\therefore \Delta_x$ = area of projection of Δ ABC on yz-plane

$$=\frac{1}{2}\left|\begin{vmatrix}2 & 3 & 1\\ -1 & 1 & 1\\ 2 & -4 & 1\end{vmatrix}\right|=\frac{21}{2}\text{sq. unit}$$

Similarly, the projection of A, B and C on zx and xy-planes are (1, 0, 3), (2, 0, 1), (1, 0, -4) and (1, 2, 0), (2, -1, 0), (1, 2, 0) respectively Also, Let Δ_y and Δ_z be teh areas of the projection of the Δ ABC on zx and xy-planes respectively.

Then, $$\Delta_y=\frac{1}{2}\left|\begin{vmatrix}1 & 3 & 1\\ 2 & 1 & 1\\ 1 & -4 & 1\end{vmatrix}\right|=\frac{7}{2}$$

and $$\Delta_z=\frac{1}{2}\left|\begin{vmatrix}1 & 2 & 1\\ 2 & -1 & 1\\ 1 & 2 & 1\end{vmatrix}\right|=0$$

$\therefore$ The required area $=\sqrt{\Delta_x^2+\Delta_y^2+\Delta_z^2}$

$$=\sqrt{\left(\frac{21}{2}\right)^2+\left(\frac{7}{2}\right)^2+(0)^2}=\frac{7}{2}\sqrt{10}\text{ sq unit}$$

$\lambda=\frac{7}{2}\sqrt{10}$ sq unit $\Rightarrow\ \dfrac{2\lambda}{\sqrt{10}}=7$

70. Plane must pass through

$\left(\dfrac{1-3}{2},\dfrac{5+1}{2},\dfrac{7-1}{2}\right)$ or $(-1,3,3)$

$\Rightarrow -1+3+2\times 3=\lambda\Rightarrow\lambda=8.$

71. The d.r's of the line are given by

$l-2m+2n=0, l+3n=0\Rightarrow\dfrac{\lambda}{6}=\dfrac{m}{1}=\dfrac{n}{-2}$

Taking $y=0$, we get

$x+2z=4, x+3z=11\Rightarrow x=-10, z=7$

The line is $\dfrac{x+10}{6}=\dfrac{y}{1}=\dfrac{z-7}{-2}$

$b=6i+j-2k$

Distance of C from the line is $\dfrac{|\overline{AC}\times\bar{b}|}{|\bar{b}|}$

$$=\frac{|(13i-2k)\times(6i+j-2k)|}{\sqrt{41}}=\frac{|2i+14j+13k|}{\sqrt{41}}=\frac{\sqrt{369}}{\sqrt{41}}=3$$

72. Any point on the first line in symmetrical form is $(3r-4, 5r-6, -2r+1)$. If the lines are coplanar, this point must lie on both the planes which determine the second line.

$$\Rightarrow\ 3(3r-4)-2(5r-6)-2r+1+5=0 \quad \ldots\text{(i)}$$

and $2(3r-4)+3(5r-6)+4(-2r+1)-k=0$ ..(ii)

From Eq. (i) we get , $r=2$

Now substituting $r=2$ in Eq. (ii), then $k=4$

73 The equations of y-axis are $\dfrac{x}{0}=\dfrac{y}{1}=\dfrac{z}{0}$,

Any point N on y-axis is (0, r, 0) ….(i)

The direction cosines of the line PN

are 0–4, r–3, 0 – 5 ie, –4, r–3, –5 ….(ii)

Let N be the foot of the perpendicular from P to y-axis, then PN is ^ to the y-axis whose direction cosines are 0, 1, 0 and so from Eq. (ii), we have

$$-0.(-4)+1.(r-3)+0.(-5)=0 \qquad \Rightarrow r=3$$

From Eq. (i) the coordinates of N are (0, 3, 0)

Required distance = PN

$$=\sqrt{(4-0)^2+(3-3)^2+(5-0)^2}=\sqrt{41}\text{ unit}$$

$$\lambda=\sqrt{41}\Rightarrow\frac{5\lambda^2}{41}=5$$

74. The lines are $\dfrac{x}{0}=\dfrac{y-b}{b}=\dfrac{z}{-c}$ and $\dfrac{x-a}{a}=\dfrac{y}{0}=\dfrac{z}{c}$

$$(bj-ck)\times(ai+ck)=bci-caj-abk$$

$$\bar{n}=\frac{bci-caj-abk}{\sqrt{b^2c^2+c^2a^2+a^2b^2}}$$

The points on the lines are $\bar{a}=bj, c=ai$

$$\Rightarrow c-a=ai-bj\Rightarrow L=\vec{n}.(\vec{c}-\vec{a})=\frac{2abc}{\sqrt{a^2b^2+b^2c^2+c^2a^2}}$$

$$\therefore L^2\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)=4$$